



UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

Communication Skills – II (EN3106)

Lab 04 - LaTeX

1. Create a new file called main.tex
2. Use the article document class with a4 paper and 12pt options using
`\documentclass[a4paper,12pt]{article}`
3. Use a graphic package to include figures.
`\usepackage{graphicx}`
4. The title of your report is **Ocean Pollution in Sri Lanka** and add Author name and today's date.
`\title{Ocean Pollution in Sri Lanka}`
`\author{Author's Name}`
`\begin{document}`
`\maketitle`
5. Write the abstract of the document as below:
`\begin{abstract}`
Sri Lanka faces an unprecedented pollution crisis as waves of plastic waste from a burning container ship hit the coast and threaten to devastate the local environment. In this paper we review solutions to this problem.
`\end{abstract}`
6. Insert a **table of contents**, **list of figures** and a **list of tables** for the document
`\tableofcontents`
`\listoffigures`
`\listoftables`
`\clearpage`

7. Write the **Introduction** to the report

```
\section{Introduction}
```

Thousands of navy sailors have been using mechanical diggers at beaches to scoop up tonnes of tiny plastic granules that have come from the Singapore-registered MV X-Press Pearl, which has been smouldering on the horizon for 10 days.

Sri Lanka's Marine Protection Authority (MEPA) said the microplastic pollution could cause years of ecological damage to the Indian Ocean island.

8. Name the next section **Background** and its subsection, **Stories**

```
\section{Background}
```

This section outlines the background of the problem that is Ocean Pollution in Sri Lanka, that accelerated drastically within the last few months with ramifications that will be seen for years and years to come.

```
\subsection{Stories}
```

Residents in coastal areas have raised issues of the areas becoming inhabitable due to pollution and loss of income. Residents suffer major income setbacks as their main income is based on fisheries which have now been affected devastatingly by the X-Press Pearl catastrophe.

9. Insert a suitable image to the subsection Stories

A snapshot of X-Press Pearl is shown in Figure~\ref{fig:pearl}.

```
\begin{figure} [!h]
\begin{center}
\includegraphics[width=4in,height=2in]{image1}
\end{center}
\caption{X-Press Pearl Incident}
\label{fig:pearl}
\end{figure}
```

10. Begin a new section, **Affected Areas** and insert a table by citing its reference.

```
\section {Affected Areas}
```

A pollution comparison of the three coastal areas \cite{ABC} are given in Table~\ref{destable}.

```
\begin{table} [!h]
\begin{center}
\begin{tabular}{|c|c|c|c|}
\end{tabular}
\end{center}
\end{table}
```

```

\hline
Area & Coastal Pollution & Air Pollution & Water Pollution \\
\hline
Galle & No & Yes & Yes \\
Matara & No & No & Yes \\
Hambantota & No & No & No \\
\hline
\end{tabular}
\caption{Area-wise Pollution Comparison}
\label{destable}
\end{center}
\end{table}

```

11. Begin a new section, **Solutions** and insert a numbered list

```

\section{Solutions}
\begin{enumerate}
\item Solution 1
\item Solution 2
\item Solution 3
\end{enumerate}

```

12. Begin a new section, **Conclusion** and insert a math equation with a footnote

```

\section{Conclusion}
The suggested solutions can be summarized as follows. Assuming
there are  $n$  solutions, their usage matrix\footnote{with a
footnote} is:

```

```

\begin{displaymath}
\mathbf{X} =
\left( \begin{array}{ccc}
x_{11} & x_{12} & \ldots \\
x_{21} & x_{22} & \ldots \\
\vdots & \vdots & \ddots
\end{array} \right)
\end{displaymath}

```

13. Insert the bibliography

```

\bibliographystyle{plain}
\bibliography{test}

```

- a. Make the bibliography file as test.bib and insert the following:

```

@ARTICLE{ABC,
  author = {A.B.C. Fernando},
  title = {Ocean Pollution},

```

```
journal = {Wonder Books},  
year = {2015}  
}
```

14. Insert an **Appendix** with **Supplementary Readings** and **Environmental Organizations** as the sections

```
\appendix
```

```
\section{Supplementary Results}
```

```
\section{Environmental Organizations}
```

The list of recognized environmental organizations are as follows.

15. End document.

```
\end {document}
```

16. Compile by running the following in the terminal¹
pdflatex main
bibtex main
pdflatex main

¹ In case the citation doesn't appear in the pdf (but appears as [?]) then insert the command `\typeout{}` before the command `\bibliography{}`